

SOFTWARE ENGINEERING & DEVOPS LAB
Laboratory Project Report
DevOps-Based Case Study with Automated CI/CD Pipeline

Case Study: Smart Parking Slot Booking System
Student Name: MOHAMMED ABDUL AFNAN
Academic Year: 2025 - 2026
Repository: (Add your GitHub URL after push)

Chapter 1: Introduction

The Smart Parking Slot Booking System is a simple Java Maven project that demonstrates slot management.

1.1 Objectives

- Develop a simple Smart Parking Slot Booking System in Java
- Implement unit tests using JUnit
- Configure Maven for build and dependency management
- Add CI pipeline using Jenkins
- Create Dockerfile to containerize the app
- Add Ansible playbook for deployment
- Maintain source code in Git and push to GitHub

1.2 Tools and Technologies

- Java 11
- Maven 3.x
- JUnit 4.13.2
- Git
- Jenkins
- Docker
- Ansible

Chapter 2: System Development

2.1 Project Structure

- pom.xml
- src/main/java/com/smartparking/model/ParkingSlot.java
- src/main/java/com/smartparking/model/Booking.java
- src/main/java/com/smartparking/service/ParkingService.java
- src/main/java/com/smartparking/Main.java
- src/test/java/com/smartparking/service/ParkingServiceTest.java
- Dockerfile
- Jenkinsfile
- deploy.yml

2.2 Class Descriptions

- ParkingSlot: Slot entity with id, type, and availability
- Booking: Booking entity with booking id, vehicle number, and slot
- ParkingService: Handles slot CRUD, booking, cancellation
- Main: Demonstrates simple execution flow
- ParkingServiceTest: Contains unit tests

Chapter 3: UML Diagrams

UML source files are available in docs/uml:

- usecase.puml
- class.puml
- sequence.puml
- deployment.puml

Chapter 4: Bug Identification & Test Cases

Detailed table is available in: docs/bugs-and-testcases.md

Chapter 5: Version Control - Git

5.1 Git Commands Used

```
bash
git init
git add .
git commit -m "Initial commit: Smart Parking Slot Booking System"
git branch -M main
git remote add origin <your-github-repo-url>
git push -u origin main
```

5.2 Expected Output

- Repository initialized
- Files staged and committed
- Code pushed to GitHub main branch

Chapter 6: Build Tool - Maven

Use this command:

```
bash
mvn clean compile test package
```

Expected result:

- Project compiles
- JUnit tests execute
- JAR generated in target/

Chapter 7: Continuous Integration - Jenkins

Jenkins pipeline stages in Jenkinsfile:

1. Clone
2. Build
3. Test
4. Package
5. Docker Build
6. Docker Run

Chapter 8: Configuration Management - Ansible

Ansible playbook file: deploy.yml

Use:
bash
ansible-playbook deploy.yml

Chapter 9: Containerization - Docker

Build and run:
bash
docker build -t smart-parking-slot-booking-system:latest .
docker run -d --name smart-parking-app smart-parking-slot-booking-system:latest

Chapter 10: Conclusion

This project demonstrates a complete yet simple DevOps workflow for a Java-based Smart Parking S